Practice: answer key

Open after the attempt. Equivalent accurate wording is accepted.

Component	Useful function	Useful source
Carbohydrate	Energy for respiration	Rice, bread, potatoes
Protein	Growth and repair materials	Beans, lentils, eggs, fish
Fat / lipid	Energy store; insulation; vitamin absorption	Oil, nuts, seeds
Vitamins	Specific processes in small amounts	C: peppers; A precursor: carrots
Minerals	Specific processes, e.g. iron for haemoglobin	Iron: lentils; calcium: dairy/fortified alternatives
Fibre	Helps food move through the gut	Wholegrains, fruit, vegetables
Water	Medium for many reactions; transport; cooling	Drinking water; water in foods

Four deficiency effects

Iron → iron-deficiency anaemia; vitamin A → impaired night vision/vision loss, severe prolonged deficiency can cause blindness; vitamin C → scurvy; vitamin D → rickets in children.

Guided practice

Harbour: iron below needs → possible iron-deficiency anaemia → lentils because they supply iron.
Garden: vitamin C below needs → possible scurvy → peppers because they supply C. Both have a nutrient gap despite adequate energy.

Original four-mark answer and scheme

A four-mark model. Energy needs are met, so the record does not show starvation. Vitamin A is below needs, so this is a nutrient deficiency and malnutrition. Severe prolonged vitamin A deficiency can impair vision or cause blindness. The supplied carrots provide beta-carotene, which the body can convert to vitamin A, helping address the stated gap.

Important distinctions. Original EduOS formative question, not an exam-board paper. Accept equivalent accurate wording. Do not require or infer a clinical diagnosis; “can cause” or “risk” is appropriate. A bare “eat more” earns no change mark. The given one-nutrient gap does not prove every other nutrient is missing.

Four marking points. Energy needs are met, so not starvation (1). · Vitamin A below needs is a nutrient deficiency/imbalance, therefore malnutrition (1). · Possible impaired vision/night blindness/blindness from vitamin A deficiency (1). · A suitable source with explanation: supplied carrots provide beta-carotene converted to vitamin A, so address the gap (1).

Review criteria

Every component connects to a useful function/source. Every deficiency links to its correct possible effect. The change addresses the stated nutrient gap. Keep the first response, substantive revision and different unaided response distinct. This key contains no protected fresh-case questions or answers.